

R. Brown
April 90

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Sixteenth Annual In-House Convention

Department of Psychology

Dalhousie University

25-26 April 1990

Room 4258/63

Sixteenth Annual In-House Convention

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Schedule

Wednesday, 25 April

9:45-10:00 Opening Remarks: R. Brown

10:00-11:00 Session I. Chair: V. M. LoLordo

- 1) D. P. Phillips, W. Danilchuk, J. Ryon and J.C. Fentress
- 2) J. Goodwin
- 3) M. Farmer
- 4) R. Brown

COFFEE BREAK

11:20-12:25 Session II. Chair: S. Nakajima

- 5) T. Juckes and J. Barresi
- 6) C. Moore and J. Barresi
- 7) H. Abe and B. Rusak
- 8) P. Dunham and A. Hurshman

LUNCH BREAK

1:30-2:45 Session III. Chair: R. Rodger

- 9) M. G. Yoon and M. Yoon
- 10) H. Schellinck and R. Brown
- 11) P. McGrath
- 12) S. Wright
- 13) J. L. Potvin and S. Nakajima

COFFEE BREAK

3:05-4:25

Session IV. Chair: D. Mitchell

- 14) S. R. Shaw
- 15) B. Frankland and A. Cohen
- 16) C. J. Beaver and D. E. Mitchell
- 17) J. Durup Connors
- 18) A. Cohen, B. Frankland, M. Lamoureux and D. Dunphy

BEER IN LOUNGE

Thursday, 26 April

9:15-10:30

Session V. Chair: M. Legerstee

- 19) J. C. Fentress
- 20) J. A. Goldman, E. M. Cosica and J. C. Fentress
- 21) B. D'Entremont
- 22) R. Braha and P. McGrath
- 23) B. Rusak, H. A. Robertson, W. Wisden and S. P. Hunt

COFFEE BREAK

10:50-11:55

Session VI. Chair: J. Clark

- 24) D. P. Phillips
- 25) V. Rouse and R. Brown
- 26) J. Barresi
- 27) G. Boyd and C. Moore

LUNCH IN LOUNGE

12:55-1:30

Session VII. Posters

- 28) A. Cohen and D. Dunphy
- 29) R. Klein and T. Jukes
- 30) C. L. Ryan and R. Brown

1:30-2:30

Session VIII. Chair: D. Williams

- 31) A. Droungas and V. M. LoLordo
- 32) A. Pontefract
- 33) V. Honig and K. Stewart
- 34) J. H. Brandstätter, I. A. Meinertzhagen and S. R. Shaw

COFFEE BREAK

2:50-3:50

Session IX. Chair: R. Ingenmey

- 35) R. Klein
- 36) N. B. O'Regan and S. Nakajima
- 37) Daphne Mitchell, M. Ozier and J. McCabe
- 38) K. G. Bina, M. Wilkinson and B. Rusak

COFFEE BREAK

4:10-5:10

Session X. Chair: S. Hall

- 39) C. Ryan and R. Brown
- 40) C. Moore, D. Furrow and L. Chiasson
- 41) A. Kingstone and R. Klein
- 42) M. G. Yoon

DINNER AND BEER IN LOUNGE

Abstracts

1. D.P. Phillips, W. Danilchuk, J. Ryon and J.C. Fentress

"Cache me if you can", the chuckling chicken cackled, or How Captive Coy Canids Cache Cadaverous Chicks.

Behaviors as (arguably) simple as leech swimming, or as (arguably) complex as mammalian walking reflect control by central motor programs: they are overt expressions of the activity of neural mechanisms which stereotype a temporal sequence of motor responses. When the sequence links distinct movements of normally unrelated body parts, there may be evidence of "synaptic rules of action". To date, the only published evidence for such rules in mammals has come from studies of grooming in laboratory rodents. As part of a much larger study, we have gathered evidence of a highly temporally structured, protracted sequence of distinct actions used by two canoid species to cache food for later retrieval. The action sequence was independent of the sex of the animal, the season in which the observations were made, and the substrate in which the food was buried. We argue that some features of the action sequence are not constrained by factors external to the animal. The action sequence may thus be the first demonstration in a large mammal, in a natural setting, of endogenous rules of action syntax.

2. Joanne Goodwin

The Role of Expressed Emotion in Relapse of Schizophrenia

Research examining the role of the emotional climate within families of schizophrenics plays in the exacerbation of schizophrenic symptomatology has been ongoing for the past 30 years. Studies in this area have generated a body of literature focusing on the degree of "expressed emotion" (EE) in families, repeatedly finding a strong relationship between high familial EE and the subsequent relapse of schizophrenic members. Despite numerous replications of this effect, the concept of EE continues to provoke controversy. Some recent studies have produced negative results; variability in approach and methodology have served to obscure the nature of the EE effect. Fundamental methodological issues involved in the EE controversy and the current status of the construct in clinical management of schizophrenia will be presented.

3. Mary Farmer

A Comparison of the Sequential Processing Abilities of Dyslexic and Normal Readers

It has been suggested by Tallal that the phonological difficulties apparent in many dyslexics are in fact symptoms of an underlying deficit in processing rapidly occurring sequential stimuli. Perception of spoken language is particularly vulnerable to such a deficit, because many of the component sounds in speech involve rapid spectral changes over time periods in the tens of milliseconds range. It is proposed to assess the rapid sequential processing ability of adolescent dyslexics, using auditory, visual, and motor sequencing tasks. The dyslexics' performance on all tasks will be compared to that of both age-matched and reading-level matched normal readers of equivalent intelligence.

4. Richard Brown

Twenty years a rat-man: Searching for a Common Thread Among the Tangled Web of My Research

My research on animal social behaviour has been organized over the last 20 years by a series of planned experiments, accidental encounters, serendipitous results and late nights in the pub. In this talk I will attempt to illustrate the common thread in a tangled web of experiments on sexual behaviour, aggression, parental behaviour, olfactory communication, ultrasonic vocalizations, sexual differentiation, hormonal stimulation, developmental differences, mate recognition, behaviour genetics and operant conditioning. The results will examine the characters, setting and, possibly the solution to the elusive question "What am I doing here?".

5. Tim Juckes and John Barresi

A "We" Mind is Bigger and Better

Individualism has produced a psychology which gives a historical and a social explanation^s of the mind. A non-individualist ontology, grounded in the social relations between individuals and mediated to individuals through social practices (e.g., language), is developed. Wittgenstein's rejection of individualism through his remarks on rules and private language are reviewed. In response to the popularity of mind-brain supervenience models, Burge's counterfactual example of "arthritis" on Earth and twin-Earth is used to show that different mental states are possible when identical brains exist. It is argued that the language—or more broadly, the social relations—of a given community must be used to determine the mental state of an individual.

6. Chris Moore and John Barresi

Theory of Mind in Autism and Psychopathy

In this paper, we suggest that two psychopathological syndromes may represent a double dissociation of two components of a theory of mind. Recent work of autism has shown that autistics do not show protodeclarative communicative acts, which have been assumed to index the initial stages of the capacity for representation. Autistics do, however, produce and comprehend protoimperatives communicative acts, which indicate the expression of emotional states of desires. Psychopaths do appear to recognize the existence of belief states in themselves and others. However, current theoretical work on psychopathy suggests that the primary problem in psychopaths is a lack of intense emotional states. This deficit prevents the psychopath from experiencing through empathy the emotional states of others and of self across time. They are thus unable to construct a concept of desire. This account predicts that there will be a subgroup of conduct disordered children who, while performing essentially normally on belief tasks, will show particular difficulty with tasks assessing the understanding of desire.

7. Hiroshi Abe and Benjamin Rusak

Revolutionary Applesauce Diet Plan: Slim Down and Get Entrained!

The circadian system of hamsters does not typically become synchronized (entrained) to daily presentations even of a highly preferred, palatable diet. We asked whether hunger would potentiate the effects of daily feeding schedules on the circadian system. When hamsters' body weights were reduced to 80-88% of free-feeding weights, many hamsters showed clear anticipatory increases in activity before the daily feeding times. Some hamsters also showed entrainment on their freerunning circadian rhythms to these schedules. The results suggest that in an appropriate motivational state, hamsters can be entrained by non-photic cues acting directly or indirectly on the circadian pacemaker.

8. Phil Dunham and Alan Hurshman

Precopulatory Mate-guarding in Gammarus lawrencianus

We will tell you a few things you probably don't know about the sex life of the amphipods in St. Margaret's Bay. At the minimum, this knowledge should make your next visit to tide pools at Peggy's Cove a little more interesting.

9. M. G. Yoon and Myonghae Yoon

Development of Narrative Representation

We study how normal children acquire the cognitive competence for mental representation of a story by longitudinal and cross-sectional observations of their narrative utterances, and detailed analyses of their linguistic structures and cognitive organizations. Preliminary results on the following questions will be presented: (1) At what age do children begin to produce narrative utterances which are relevant to the input story? (2) Which functional categories of the constituent propositions of the input-text do the children regard important and/or interesting to represent them in their narrative outputs? (3) At what age do the children become able to narrate the gist of the story? (4) How do children interact pragmatically with their addressees during their narration (e.g. scaffolding, presupposition, etc.)? (5) Do the children make their own interpretations (e.g. casual inference, projection of certain propositional attitudes, imputation of different mental states or perspectives to the characters in the story, etc.) of the input-text, and thus transform it in a certain systematic way in their narrative representation?

*Language + thought of Piaget.
14 yrs 10 months - dev. of mind*

10. Heather Schellinck and Richard Brown

Genetic and Microbiological Bases of Individual Odours in Rats

Both genetic and microbiological factors may influence the ability of animals to discriminate between individuals by their odours. To determine the role of each, we examined the ability of Long Evans male rats to discriminate between urine odours from the following pairs of male rats in an olfactometer: (1) outbred Long Evans, (2) MHC congenics, (3) germfree MHC congenics, (4) genetically identical individuals. Discriminable differences were present in the urine odours used in all four tasks. Analysis of the errors to a criterion of 85% correct discrimination responses revealed different learning patterns in each task indicating that genetic and microbiological differences both contribute to their individual odours of rats. When there are no genetic differences (Task 4) or no microbiological differences (Task 3) discrimination was most difficult; when both genetics and bacteria differed (Tasks 1 & 2) discrimination was easier. The easiest discrimination (Task 2) involved the presence of bacteria and a genetic difference at the major histocompatibility complex (MHC) of genes on chromosome 14.

11. Pat McGrath

Is it sane to think of pain on the plane? Answers from WHO.

Pain transmission theories such as the gate control theory have been useful in enhancing our understanding of pain and have legitimized the role of psychological factors in pain. They have not, however, provided any guidance to clinicians or researchers in how to assess pain. Because of the lack of an appropriate theoretical model pain assessment has not progressed as much as one of pain assessment that may provide a strategy for advancing pain assessment.

Measurement is Assessment

Cross-cultural expression of pain - Darwin 1872

12. S. Wright

Importance of Paternal Care for Pup Survival and Development in Monogamous vs. Polygynous Mice

In monogamous species of mammals, males usually show paternal behaviour, while in polygynous species males are not normally parental care on pup development in a monogamous species (*P. californicus*) and a polygynous species (*M. musculus*) of mouse. Since paternal care may be shown only when females need to leave the nest, we designed an experiment in which wheel running is necessary to "forage" for food. This 2 x 3 mixed factorial experiment has two levels of parental care mixed factorial (male absent vs. male present) and three levels of foraging (wheel running for food; free food plus wheel running; free food and no wheel). We measured litter weight, parental behaviour and number of revolutions run per hour in each group in both the light and dark phase of the L:D cycle. This talk will focus on species differences in pupweight and pup survival under the six conditions and discuss species differences in paternal care.

13. James L. Potvin and S. Nakajima

expectation glutamate - stim. kainic acid receptors - in fish

Neural and Behavioural Effects of Domoic Acid, an Amnesic Shellfish Toxin, in the Rat

Among those who were poisoned by shellfish containing domoic acid, 13 persons suffered from varying degrees of memory loss. To study the mechanism of the amnesic effect, domoic acid was injected into the cerebral ventricle of 53 rats. They showed spike discharges in EEG and loss of pyramidal neurons in the hippocampus. They were also impaired in learning a radial-arm maze. In the animals previously trained on the maze task, domoic acid interfered with relearning. The memory deficits following the ingestion of tainted shellfish seem to have resulted from lesions in the hippocampus produced by domoic acid.

- objection? Motor convulsions

14. S. R. Shaw

Split Personality and Identity Crisis in a Subject with Sensational Legs

If a part of a nervous system can be turned on in two quite distinct ways by the hapless experimenter, how does the onlooker decide which of these two is the proper route to activation? Suggestive comments will be made about one way to arbitrate on this dilemma, while taking a fresh look at a previously misinterpreted neural system. The results indicate provisionally that in order to arouse this subject, a properly pitched word in the ear beats the more traditional hand-on-the-knee approach.

15. Bradley W. Frankland, Annabel J. Cohen

Effects of Tonality on Accuracy and Reaction Time in an Interpolated Pitch Memory Task

Three groups of listeners (N = 12) differing in level of musical training (no, low, and high) indicated, on each trial, whether the first (standard) and last (comparison) tone of a five-tone melody were the same or in different pitch. On one-half of the trials, the standard and comparison tone were identical; on the other half the comparison was one semitone higher than the standard. The three middle (interpolated) tones of each five-tone melody formed either a major or minor triad in a key either near to or remote from that of the standard, thus providing four levels of decreasing tonality relation with the standard. Accuracy increased and reaction time decreased with increasing tonality relation between the interpolated tones and the standard, producing significant linear trends for both dependent measures. Both measures demonstrated flatter functions for the group which had no formal musical training. These results are consistent with the application of Krumhansl's tonal hierarchy for musically trained listeners. The study also confirms the use of reaction time for measuring the mental representation of melodic structure. (Work supported by NSERC).

16. C. J. Beaver and D. E. Mitchell

Cortical Areas Mediating Behavioural Recovery Following Early Periods of Monocular Deprivation in Kittens

Previous work in the laboratory has shown that following termination of 6 weeks of reverse occlusion of kittens that had been monocularly deprived until 6 weeks of age there is a surprising change in the visual acuities of both eyes. Although the visual acuity of the initially deprived eye improves substantially during the period the vision of the eye deteriorates rapidly while that of the initially non-deprived eye regains almost normal acuity. As of yet, the mechanism mediating these changes is still unknown. Lesions of areas 17 and 18 have been carried out to determine the location of the mechanism. Outcomes and their meaning will be discussed.

17. Josette Durup

Visual Attention Bias in Trait Anxiety

In the last decade there has been a blossoming interest by some cognitive psychologists/researchers investigating information processing models in various affective states. The studies carried out so far have been rather convincing at showing how individuals with various affective states differ from each other and from normals in the way they process information. The differences have been observed at various stages of the information processing continuum, be it at the perceptual, interpretive or recall/response level. This line of research can assist in our understanding of these disorders and may have implications in their treatment.

I will talk about an empirical research to investigate biases at the visual attentional/perceptual stage of information processing in high and low trait anxious individuals.

18. A. Cohen, B. Frankland, M. Lamoureux, D. Dunphy

Items and Orders in Melodic Tracking: The NeXT Step

Listeners indicated the serial order of 8 tones in 36 sequences representing all combinations of 4 8-note scales and 9 serial orders. Accuracy was higher for scales with wider frequency ranges, for serial order with simpler contour changes, for sequences in which scale did not vary from trial to trial and for listeners with musical training. These effects are discussed from the perspectives of concatenative versus rule-governed accounts of sequential learning. The experiment also represents successful application of the NeXT computer in a serial tracking task.

19. John C. Fentress

Look Mom! No Malice in Cognitiveland

One of the interesting but poorly explored questions in ethology is how we code behaviour. What features do we attend to, what are our biases, and how do observation profiles emerge during development? In this talk I offer some preliminary explorations. First, a brief comment on work being done with Rita Anderson at Memorial (wolf watching). Secondly, confessions from sabbatical voyeurism at Oregon ("spot" watching). With Mike Posner and Mary Rothbart I scored video tapes of adult and infant attentional movements. Preliminary analyses of "attention runs" suggest parallels to ethological evaluations of patterned animal performance. If McDonnell-Pew cooperates, Posner, Rothbart and I plan to extend this research line.

20. Jill A. Goldman, Elizabeth M. Coscia, and John C. Fentress

Wolf ear movements: A Measure of Maternal Responsiveness

We examined communication between a mother wolf and her pups in the den. Previous work has shown that although neonatal pups emit a diverse repertoire of sounds during this period, the mother rarely responds by vocalizing (Coscia, 1989, unpublished). These findings prompted our investigation of pup vocalizations which might elicit other forms of behavioural response. A 180-hour videotape database gave us the opportunity to observe mother and pups' behavioural and vocal activity during the first 5 postnatal weeks. Mother's responses were coded for both ear and body movements. The data suggest that most pup vocalizations do not elicit an overt behavioural response. However, the majority of mother's ear movements do occur while the pups are active and vocal. The latter has given us reason to believe that the mother's ear movements represent a form of monitoring which is used to decipher which pup vocalizations are acted upon.

21. Barbara D'Entremont

Concept Development in Preschoolers

It has been shown that preschoolers organize their concepts thematically (e.g. dog-bone), but the presence of a novel label biases them to look for a taxonomic relationship (e.g. dog-dog). Fifty-six three-year-old children were tested to determine if a general labeling pretreatment would also bias them to respond taxonomically. A picture was placed in front of the child and they were asked to "find another one that is the same as this one" from a taxonomic, thematic or irrelevant choice. In the novel label conditions, the child was asked to "find another X that is the same as this X", where "X" was a nonsense word. The labeling pretreatment consisted of asking the children to label various pictures. Four groups were formed (pretreatment-label/—no label, no pretreatment-label/—no label). Results showed the novel label group made more taxonomic choices than the no label group only when there was no pretreatment. When there was a pretreatment, the no label group did as well as the novel label group. This supports the argument that language provides a taxonomic constraint on children's interpretation of word meanings.

22. Richard Braha and Pat McGrath

Methods for doing literature reviews.

No Pain No Gain: Looking back at Pain. How good were the gains?

Treatment strategies for pain abound. The efficacy of these strategies is well documented: major review papers tell us this! Unfortunately, most of the reviews are flawed. In ten minutes, I will (1) review problems with the way researchers synthesize knowledge, (2) revive the adage, "garbage in, garbage out"—as it applies to methods of data collapsing, and (3) present a proposal to implement what I have facetiously called, "A Best Evidence Critical Appraisal Meta-Analytic Synthesis"***

*** portions of this label are stolen from others

23. B. Rusak, H.A. Robertson, W. Wisden and S.P. Hunt

Getting into your Genes: It's not always worth the Fos

Last year, we found that light stimuli cause increased levels of the Fos protein to appear in cells in the hamster circadian pacemaker, the suprachiasmatic nucleus. We have since found that this increase results from altered gene expression in response to light input. This altered gene expression occurs at the time of day when light can shift the circadian clock, but not at other times of day when light is ineffective in shifting the clock. The results imply that altered gene expression is intimately linked to the mechanism that resets the circadian clock. These findings may provide a novel approach to the identification of the genes responsible for running the circadian pacemaker.

24. D. P. Phillips

Timing is Transient Stimulus Events by Cortical Auditory Neurons in the Cat

My lab is engaged in a long-term study of the temporal grain with which the spike activity of auditory cortical neurons can represent the time structure of sounds. Last year, we talked about why these cells are unable to encode, in spike times, stimulus periodicities in excess of about 100 Hz. In this year's presentation, we show that in response to transient stimuli, many cortical cells discharge spikes with a precision in the range from a few hundred microseconds to a few milliseconds. We show that this response precision can support a temporal resolution in the range from about a millisecond to a few tens of milliseconds. Interestingly, this is the grain of temporal processing lost after lesions to the homologous region in man.

25. Valerie Rouse and Richard Brown

The Ability of the Rat to Discriminate Human Urine Odours

Genetic differences at the major histocompatibility complex (MHC) contribute to individual differences in urine odours of both rats and mice which can be detected in discrimination tasks. Discrimination experiments with dogs have suggested a genetic contribution to human body odours. Operant conditioning techniques were used to determine the ability of rats to discriminate human urine odours from pairs of boys which differed in genetic relatedness: unrelated, brothers, fraternal twins, identical twins. All animals were able to discriminate between the two odours on all tasks. Analyses revealed no difference in task difficulty. It was concluded that rats were not attending to genetic cues in the odours but were relearning the discrimination tasks each day based on urine concentration, metabolites in the urine or other variables.

26. John Barresi

Persona Grata: Fragments for a History of the Concept of Person

The criteria by which personhood is ascribed to an object or being have changed dramatically from primitive through civilized cultures. In the present paper I can only highlight some of the more interesting transformations and try to place them in their historical context.

27. Gisele Boyd and Chris Moore

The Significance of Maternal Feedback on Children's Linguistic Development

To date, the role of the linguistic environment in facilitating language acquisition is unclear. In an attempt to resolve this issue, the present study examined one particular aspect of input, maternal sensitivity to the grammaticality of the child's utterance. The purpose of the present study was two-fold. The first issue was to investigate whether mothers provided differential feedback towards their child's grammatical and ungrammatical utterances. The second issue was whether such maternal feedback had any influence on the child's subsequent utterance. Nineteen children were observed with their mothers in naturalistic conditions at two age levels, 24 months of age and 36 months of age. Transcribed conversations between the mother and child were analyzed, and the results indicated that mothers were indeed sensitive to the grammaticality of the child's utterance. In addition, the results suggest that the children were influenced by their mother's immediate feedback and often reformulated their subsequent utterance to reflect this understanding.

28. Annabel Cohen and Debora Dunphy (Poster)

Musical and Visual Processing in Film

The present study examines whether processing of musical soundtracks exploits general of modality-specific resources. Subjects received either a film + music, film-only or music-only presentation for 20 min and subsequently completed a respective 25-item recognition test. In the film + music test, recognition excerpts were either unchanged, had a new musical track, had a new visual track, were mismatched visual and musical tracks, or were entirely novel. Music-only and film-only tests used one modality from the film + music test tape. Subjects distinguished old from new material in both modalities in both single- and double- modality response patterns confirmed that processing film music did not compete for visual resources. Intermodal effects were also apparent.

29. Raymond Klein and Tim Juckes (Poster)

Can Auditory Frequency Control the Direction of Visual Attention?

Tones of fixed frequency (250, 650 and 2500Hz) or frequency glides (900-400Hz and 700-1200Hz) emanating from straight ahead had no effect on reaction time to luminance targets above or below fixation. This suggests that the phenomenal correspondance between pitch and location does not involve reflexive control of visual orienting mechanisms. Recent studies manipulated the relationship between the direction of the frequency glides and the likelihood of the visual target appearing above or below fixation. The results indicate that subjects can shift their visual attention in the congruent direction within about 500ms while incongruent shifts are not observed with 990ms.

30. C.L. Ryan and R. E. Brown (Poster)

The Effects of Yohimbine on the Copulatory Behaviour of Older Male Rats

The present study was designed to determine whether age related deficits in copulatory behaviour in male rats could be reversed by the alpha 2 antagonist, yohimbine, a reported aphrodisiac.

An acute dose of yohimbine was administered to eight 15 month old male rats, who had previously demonstrated deficits in copulatory behaviour compared to that of eight 6-8 month old male rats. The copulatory behaviour was assessed in a 20 minute trial with a receptive female.

Yohimbine altered copulatory performance and tended to shift the behaviour of the older males towards that of the younger age performance. The drug also caused a decrease in 22kHz post ejaculatory vocalizations. This may reflect changing arousal levels or a specific effect of the drug on ultrasonic vocalizations.

31. Anastasia Droungas and Vincent M. LoLordo

In Search of Parametres that Yield a Pure Explicitly Unpaired Safety Signal

As a result of Pavlovian Explicitly Unpaired (EUP) conditioning, whereby a CS and shock US occur on separate trials, the CS becomes a safety signal. We have shown that a CS trained with a variant of not the EUP procedure (a brief CS singly alternated with shock with a short intertrial interval) functioned both as a safety signal AND as a fearful stimulus. We argued that the CS functioned as a safety signal because, during conditioning, it was negatively correlated with shock; and as fearful stimulus because, during conditioning, it was associated with the fearful conditioning context. The current experiment explored parametric variations of EUP aiming at establishing conditions which will yield a "pure" EUP safety signal.

32. Amanda Pontefract

In Search of Inhibition of Return

Events in the visual field often attract attention as indicated by a nearly immediate facilitation for the processing of nearby stimuli. In contrast, Posner and Cohen discovered a delayed inhibitory effect which begins about 300 ms after a visual event if attention is not maintained at the location of that event. This phenomenon, known as "inhibition of return", serves to decrease the probability that attention returns to a previously attended location. This talk will focus on the experimental conditions necessary for producing this inhibitory effect.

33. Vern Honig and Karen Stewart

Numerosity as a Perceptual Continuum: How to get by Without Actually Counting

Cognitive and developmental psychologists are preoccupied with counting, especially of smaller numbers, perhaps up to the "Magical Number Seven". This may be due to the fact that people can "subcept" such numbers and count them even after a brief presentation. As a result, many animal psychologists concentrate on counting with small numbers as well. We have data that show that pigeons can discriminate among small numbers of elements in the way that they typically do among different values on other sensory continua: They provide orderly gradients of discrimination, for example, with 2+, 3+, 4+, 5-, 6- and 7-. They cannot however, respond differentially to adjacent small numerosity values, such as 2+, 3-, 4+ and 5-.

34. J.H. Brandstätter, I. A. Meinertzhagen, S. R. Shaw

Floating Synapses

In a continuation of last year's project—the disassembly of afferent photoreceptor synapses following photo-degeneration—we have examined the effects of the loss of receptor terminals on a second major class of synapses in the lamina of the fly's eye, the feedback synapses of one of the monopolar cells, L2, back upon the receptor terminal.

A first and for us, very surprising finding was the presence of "floating synapses", presynaptic ribbons separated from their membrane sites by invagination to form an organelle that is free floating in the cytoplasm of the L2 axon.

These floating synapses seem to be a degradation product of feedback synapses. Despite this loss, we found no decrease in synaptic frequency, but an increase attaining a twofold value of 8d, suggesting the occurrence of reactive synaptogenesis.

35. Ray Klein

Orienting of Visual Attention

Perception of a visual scene can be spatially selective in two senses. First, overt adjustments of gaze direction can be made to control which regions of the visual field are processed by the sensitive fovea and its associated neural machinery. Second, covert adjustments can be made in which specific objects or regions are selected (in the absence of gaze changes) for preferential treatment. Overt and covert visual orienting can be controlled exogenously by features of the visual scene as well as endogenously by the observer's expectations and intentions. Imagine the 2x2 matrix implied by this classification. In this talk I will concentrate on the cells representing endogenous and exogenous control are quite different. This might be interpreted to imply that two different attentional systems are involved. Alternatively, a single attentional system might bring different resources or strategies to bear depending on the nature of the control signal.

~~36. Neil D. O'Regan and S. Nakajima~~

36. Noel B. O'Regan and S. Nakajima

Reinforcing Characteristics of Brain Stimulation Altered by Receptor—Specific Dopamine Agonists and Antagonists

The involvement of dopaminergic system to the generation of the reinforcing effect of brain stimulation was examined. Eight rats were trained to press a bar for hypothalamic stimulation and injected intraperitoneally with agonists and antagonists selective to either D1 or D2 subtypes of dopamine receptors. Quinpirole (D2 agonist) reduced the threshold for self-stimulation while raclopride (D2 antagonist) and SCH 23390 (D1 antagonist) increased it. SKF 38393 (D1 agonist) and physiological saline solution had no effect. These results suggest a critical involvement of dopaminergic system with D2 as well as D1 receptors in the reinforcing effect of brain stimulation.

37. D. Mitchell, M. Ozier, J. McCabe

Recognition of Videotape Sports: Coaches and Players Compared

Expert and novice coaches and players were presented 5-second samples of a ten-minute video excerpt of a basketball game they had seen, along with samples taken from elsewhere in the same game. The samples were of three types: moving samples, still photos of the video material, or still photos taken from a view point different to the video. The subjects' task was to indicate whether the sample had occurred in the ten-minute video or not. The results indicated that coaches were more able to recognize the samples than were the players, and that three types of display differed in difficulty, the moving displays being recognized by all groups better than the stills.

38. K. G. Bina, M. Wilkinson and B. Rusak

Daily Variation in Muscarinic Binding Sites in the Cortex and Hypothalamus of Golden Hamsters

Circadian variation in the binding of (^3H)N-methyl scopolamine ((^3H)NMS), a hydrophilic muscarinic receptor antagonist was studied in the hypothalamus and cortex of the Golden Hamster (*Mesocricetus auratus*) using a novel micropunch technique which selectively detects only cell surface receptors. Intact and suprachiasmatic nucleus ablated (SCNX) animals maintained in light dark cycle (LD) 14:10 were killed every four hours starting at lights on and binding assays performed on hypothalamus and the cortex. Binding assays were also performed on intact animals maintained in constant darkness (DD) every four hours starting at circadian time (CT) 0. The intact LD animals showed a significant daily rhythm in (^3H)NMS binding in the cortex with highs occurring 12 hours after lights on and lows at lights on. No significant rhythm was detected in the hypothalamus. The rhythm in the cortex was abolished in the SCN-ablated animals and in animals maintained in DD. The data suggest that photic information mediated through the SCN or the pacemaker property of the SCN is responsible for this rhythm. Autoradiography ((^3H)NMS; 2nM, 3 weeks exposure) clearly revealed specific binding in the SCN and surrounding periventricular region.

39. Cathy Ryan and Richard Brown

Modification of Neonatal Ultrasonic Vocalizations in Rats by Adrenergic Drugs

There is some confusion regarding the role of the noradrenergic (NA) system in regulating neonatal rodent ultrasonic vocalizations (UV). We decided to systematically investigate the role of NA in the production of these infant distress vocalizations.

Hooded rat pups received a single injection of one of four adrenergic drugs: propranolol, yohimbine, prazosin, or clonidine at 4, 8, 12, or 16 days of age. With the exception of clonidine, all drugs depressed isolation induced UV. Clonidine enhanced vocalizations with its effects on UV being age-dependent. The role of the NA system in the manifestation of these findings.

40. Chris Moore, David Furrow and Lorraine Chiasson

Mental state Language in Mother—Child Communication

The acquisition of a mental state language has been assigned some significance in the development of a theory of mind. For example, Olson (1988) has argued that without means of representing such attitudes, the child would never be able to acquire an understanding of propositional attitudes. However, little is known about the conditions under which mental state language is acquired. In this paper, we report data from a longitudinal study of 19 mother-child pairs in communication when the child was 2 and 3 years of age. All occurrences of maternal and child utterances containing a mental term were coded for function. The results show that while the overall use of mental terms by the mother when the child was 2 years correlated with the child's use at 3 years, particular maternal functions appeared to be especially important. Specifically, mothers who used mental terms in open-ended directives to their children at 2 years have children who used mental terms more at 3 years. We argue that mother's use of open-ended directives serves to direct their children's thought rather than action, and that such direction is important for fostering a sensitivity on the part of the child to the existence of mental life.

41. Alan Kingstone and Raymond Klein

Express Saccades

Fischer and colleagues (e.g. Fischer and Breitmeyer, 1987; Fischer and Ramsperger, 1984) have reported that human eye movement latencies to a peripheral target are unusually brief (100ms - 120ms) when a fixation point is removed shortly before target onset. This new phenomenon is called 'express saccades'. It has also been reported that it is not the offset of the fixation stimulus per se that permits express saccades, but rather the offset of any stimulus (even a parafoveal one) that is attended. This observation appears to provide a strong link between attention and the eye movement system: attention must be disengaged before a saccade can be initiated. But the conclusion is questionable since the studies which 'manipulated' attention told subjects to attend to a peripheral stimulus without providing any objective evidence that this instruction was followed.

We report a series of experiments which confirm the express saccade phenomenon, explore variables such as temporal spatial (un)certainly which may mediate the effect, and examine the role that attention may play in its occurrence.

42. M. G. Yoon

Bilateral Ablation of the Forebrain Enhances the Optomotor Responses in Goldfish

The topographic pattern of visual projection from the eye to the brain in goldfish can be experimentally induced to undergo two types of dramatic reorganizations: (1) the foveal compression following partial excision of the optic tectal lobe, and (2) the retention of original topographic polarity by tectal tissue following reimplantation after 180° rotation. We study the functional significance of these two types of neural reorganization in the goldfish by various behavioural tests. One of the innate visual behaviours we use is the optomotor response. It is not easy to elicit a reliable optomotor response to a well localized moving stimulus in normal goldfish. About two weeks after bilateral ablation of the forebrain in goldfish, however, we find that the operated fish tend to follow the moving visual target with remarkably enhanced consistency. Thus, the forebrain seems to exert a distracting influence on the optomotor response generating mechanism in normal goldfish.